Results outline

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1 Wildchrokie

1.1 Growing season length and conditions

1. Across the three years and all species, the growing season length varied from 90 days to 173 days.
2. The year with longest growing season length on average was 2019 with 141.49 days, and the shortest was 2020 with 135.87 days.
3. The hottest spring period of the three years (MAM) was 2020 with a mean mean temperature of 9.05°C, and the coldest, 2019 with a mean mean temperature of 8.72°C.
4. The hottest summer period of the three years (JJA) was 2020 with a mean mean temperature of 23.07°C, and the coldest, 2019 with a mean mean temperature of 22.38°C.
5. The driest spring period of the three years (MAM) was 2019 with a mean mean temperature of 2.74, and the coldest, 2020 with a mean mean temperature of 1.32.
6. The driest summer period (PDSI) of the three years (JJA) was 2019 with a PDSI value of 4.02, and the coldest, 2020 with a PDSI value of -2.35.
7. The spring period with the highest precipitation of the three years (MAM) was 2018 with a mean mean temperature of 120.87, and the coldest, 2020 with a mean mean temperature of 91.77.
8. The summer period with the highest precipitation of the three years (JJA) was 2019 with a PDSI value of 115.33, and the coldest, 2020 with a PDSI value of 66.37.

year pdsi$_M AMpdsi_J JAtmeanmax_M AMtmeanmax_J JAtmeanmean_M AM$12016−3.3033333−4.040000015.2666729.13333
2.42333331.586666714.1666727.366678.783333320182.44333331.923333314.3333328.400008.900000420192.74333334.020000014
2.350000014.5666728.900009.05000062021 − 1.83666670.973333316.0000028.200009.900000720220.3633333 −
2.666666715.9333329.1000010.25000082023−2.2600000−0.300000015.9000026.9666710.083333920242.88000002.263333315.266

1.2 Ring widths

1. The ring width of our species vary from 0.39mm to 13.55mm.

1.3 Model results

Our results show similar trends in response to the thermal versus the calendar growing season, with a slightly stronger response to longer growing season compared to the increased growing degree days.

1. G. alder ($\beta = 0.277$), Y. birch ($\beta = 0.175$) and P. Birch ($\beta = 0.284$) show a weak response, but seem to grow more with more GDD, while G. birch ($\beta = 0.087$) did not respond to increased GDD (Figure 2).

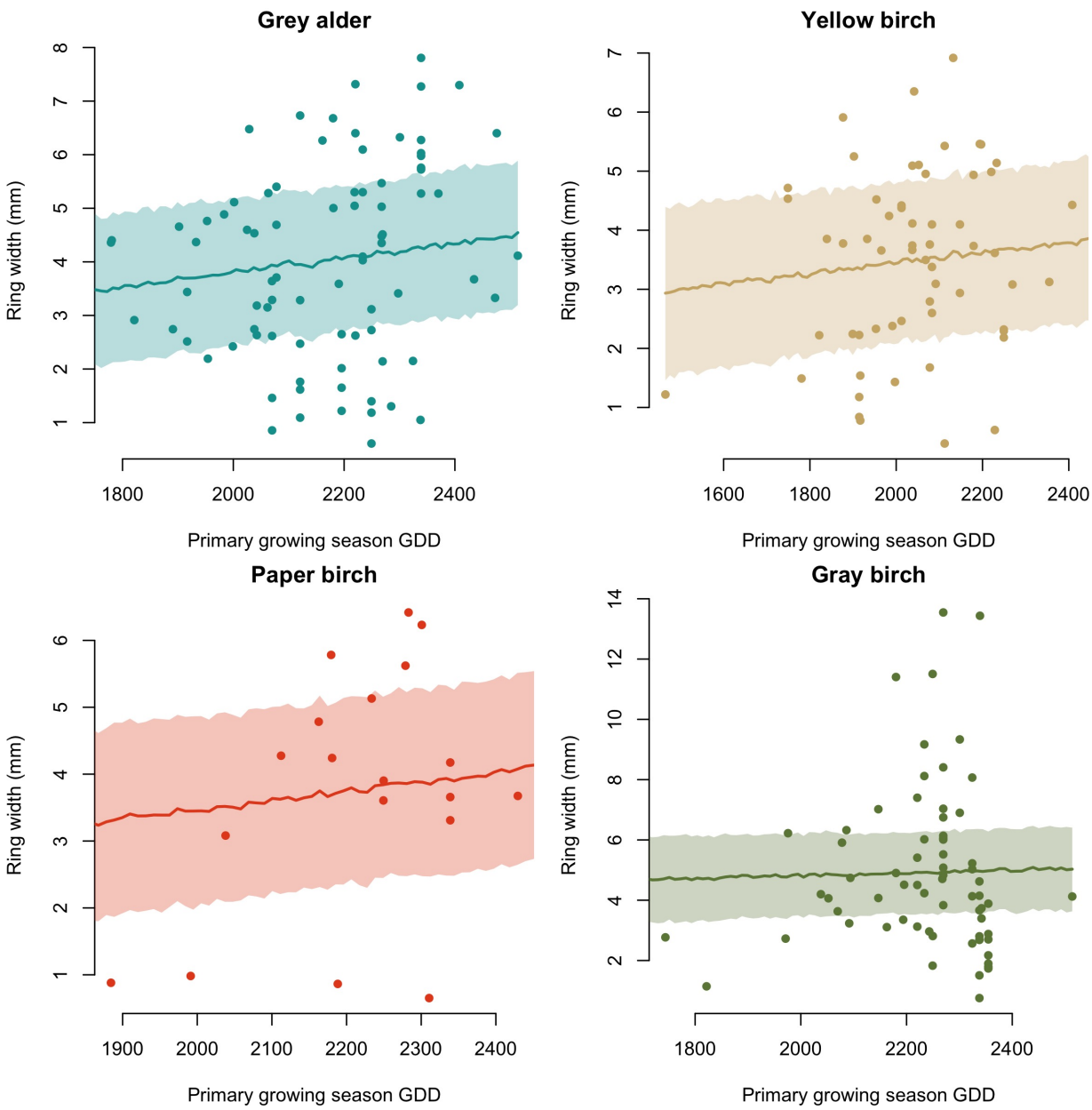

Figure 1: Ring width responses to growing degree days for the primary growing season. The shaded areas represent the 50% confidence intervals. The dots represent the empirical data

2. G. alder ($\beta = 0.374$) is the only species for which its 90% intervals don't overlap with zero. Y. birch ($\beta = 0.19$) and P. Birch ($\beta = 0.327$) show a weaker response, but seem to grow more in response to longer seasons, while G. birch ($\beta = 0.017$) did not respond to the GSL (Figure 3).

3. For the provenance with the GDD model, we detected a weak growth difference between G. College (α_{site} = -0.379), S. Hypolyte (α_{site} = -0.07) or W. Mountains (α_{site} = 0.471). Only HF grew less (α_{site} = = -0.768) (Figure 4)

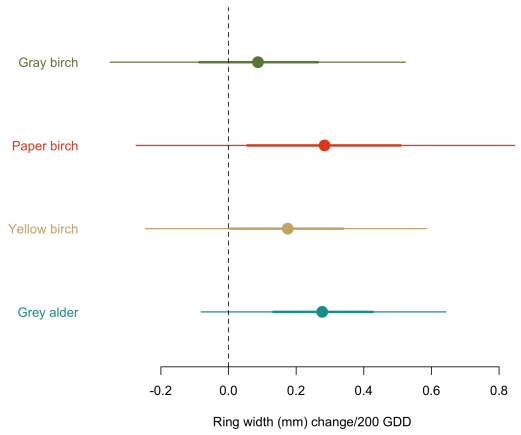

Figure 2: Slope values across the different species for GDD

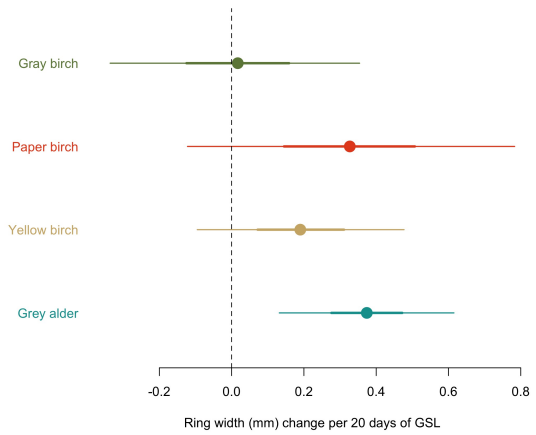

Figure 3: Slope values across the different species for GDD

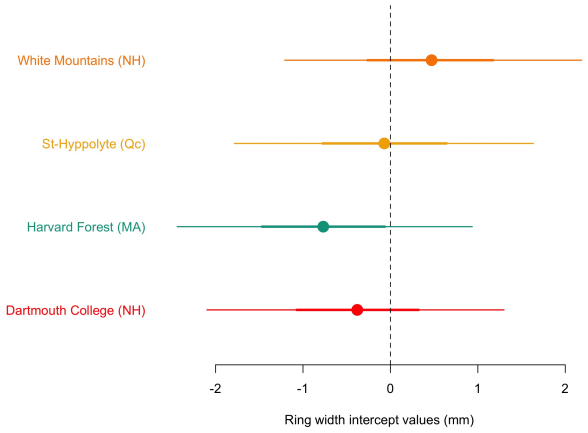

Figure 4: Site intercept values

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Supplemental results

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